DSC 40B Discussion 2 Worksheet Solutions

Friday, April 10, 2026

Imagine you are a tutor, and you need to explain something to someone who has only a vague understanding of what's going on. Do your best to get the point across. **ALTERNATE**.

If you're not sure, ask your peer if they know how to explain it, then REPEAT the explanation back. Remember, the point of the exercise is to **vocalize** your thoughts.

If you are both unsure, make sure to come to/tutor for help! Do not use ChatGPT or any other LLM! The point of discussion is to prepare you for the exams.

Have one person in the pair open the Discussion 1 Gradescope assignment. After both you **and** your partner finish a problem **and explain it**, answer the corresponding multiple choice/short answer questions for both versions.. You will receive instant feedback on the questions, whether you are correct or not. If your answer is incorrect, please go back to the question to understand it and feel free to ask tutors about anything you are confused about. Once you have finished the entire worksheet or discussion is ending, turn in your assignment and add your partner to the submission.

Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. What are you looking forward to this quarter? (Do not spend more than 2 minutes!)

Questions

Question 1:

First Person:

True or False: If algorithm A runs in $\Theta(2^n)$ and algorithm B runs in $\Theta(3^n)$, then there exists an input size n such that algorithm A completes faster than algorithm B. Explain your reasoning.

True. By the definition of asymptotic, 2^n grows asymptotically slower than 3^n, so there must exist some input size n beyond which algorithm A's runtime becomes smaller than algorithm B's. Therefore, at least one such n exists where A completes faster than B.

Second Person:

True or False: For all $f(n) = \Theta(2^n)$ and $g(n) = \Theta(3^n)$, we have $g(n) > f(n)$ for all n. Explain your reasoning.

False. f(n) could have some very large constants, causing it to be slower than g(n) for smaller inputs of n.

Question 2:

Second Person:

Sort the following by asymptotic growth rate, from smallest to largest. Explain your reasoning.

- $n!$
- 10000000^n
- $log(n^{100})$
- n^2

log(n^100) < n^2 < 1000000^n < n!

First Person:

Sort the following by asymptotic growth rate, from smallest to largest. Explain your reasoning.

- 1.1^n
- $10000000n$
- $log(n!)$
- $(log(n))^2$

(log(n))^2 < log(n!) < 1000000n < 1.1^n

Question 3:

First Person:

Using the **formal** definition of Big-Theta, find and prove the runtime of the expression below. Show your work.

$$f(n) = 4n^2 - 5n + 7$$

$\Theta(n^2)$

$4n^2 - 5n + 7 \leq 4n^2 - 5n + 7 \leq 4n^2 - 5n + 7$

$\rightarrow 4n^2 - 5n \leq 4n^2 - 5n + 7 \leq 4n^2 + 7$

$\rightarrow 3n^2 + (n^2 - 5n) \leq 4n^2 - 5n + 7 \leq 4n^2 + 7n^2$

$\rightarrow 3n^2 \leq 4n^2 - 5n + 7 \leq 11n^2,\ for\ n \geq 5$

$\rightarrow f(n)\ has\ Big - Theta\ runtime\ of\ \Theta(n^2)$

Second Person:

Using the formal definition of Big-Theta, find and prove the runtime of the expression below. Show your work.

$$f(n) = 2n^2 + 4n - 9$$

$\Theta(n^2)$

$2n^2 + 4n - 9 \leq 2n^2 + 4n - 9 \leq 2n^2 + 4n - 9$

$\rightarrow 2n^2 + 3n + (n - 9) \leq 2n^2 + 4n - 9 \leq 2n^2 + 4n^2$

$\rightarrow 2n^2 \leq 2n^2 + 4n - 9 \leq 6n^2$, for $n \geq 9$

$\rightarrow f(n)\ has\ Big - Theta\ runtime\ of\ \Theta(n^2)$

Question 4:

Second Person:

Which of the following asymptotic bounds are true? Explain your reasoning.

- [x] $\cancel{nlog(n) = O(n^2)}$
- [ ] $(\frac{1}{n})^3 = O((\frac{1}{n})^4)$
- [x] $\cancel{n^{1/2} = \Omega(n^{1/3})}$

First Person:

Which of the following asymptotic bounds are true? Explain your reasoning.

- [x] $\cancel{(log(n))^2 = O(nlog(n))}$
- [x] $\cancel{e^n = O(n!)}$
- [ ] $log(n) = \Omega(\sqrt{n})$

Question 5:

First Person:

What is the best and worst case time complexity of the code below? Explain your reasoning.

```
def runtime_A(nums):
    i = 1
    while i < len(nums):
        i = i * 2
        if nums[i] == i:
            return "What a coincidence!"
    return i
```

Best case: constant, Worst case: log(n)

Second Person:

What is the **best** and **worst** case time complexity of the code below? Explain your reasoning.

```
def runtime_B(nums):
    for i in range(len(nums)**2):
        if nums[i] == 40:
            return "I love 40B!"
        print("working...")
    return i
```

Best case: constant, Worst case: n^2

Question 6:

Second Person:

What is the **best** and **worst** case time complexity of the code below? Explain your reasoning.

```
def runtime_B(nums):
    for i in range(len(nums)):
        if nums[i] == 4:
            for j in range(len(nums)):
                print("I love 40B!")
            return "So much!"
    return "Cool!"
```

Best case: n, Worst case: n

First Person:

What is the best and worst case time complexity of the code below? Explain your reasoning.

```
def runtime_A(nums):
    for i in range(len(nums)):
        if nums[i] == 4:
            for j in range(len(nums)):
                print("I love 40B!")
        else:
            return "So much!"
    return "Cool!"
```

Best case: constant, Worst case: n^2

Question 7:

First Person:

Let $f(n) = 12log_2(3^{n^2-2n} + 2^{log(n)} - 10n^2 - log_3 n)$. Which of the following asymptotic bounds on f are true?

- [x] ~~$f(n) = \Omega(1)$~~
- [ ] $f(n) = O(1)$
- [ ] $f(n) = O(log(n))$
- [x] ~~$f(n) = \Omega(log(n))$~~
- [ ] $f(n) = \Theta(log(n))$
- [ ] $f(n) = O(n)$
- [x] ~~$f(n) = O(e^n)$~~
- [ ] $f(n) = \Theta(n)$
- [x] ~~$f(n) = \Theta(n^2)$~~

Second Person:

Let $f(n) = 5log_2(3^{nlog(n)\ -2n} + 2^{(log(n))^2} - n^5 - log_3 4n)$. Which of the following asymptotic bounds on f are true?

- [x] ~~$f(n) = \Omega(1)$~~
- [ ] $f(n) = O(1)$
- [ ] $f(n) = O(log(n))$
- [x] ~~$f(n) = \Omega(log(n))$~~
- [ ] $f(n) = \Theta(log(n))$
- [ ] $f(n) = O(n)$
- [x] ~~$f(n) = O(e^n)$~~
- [ ] $f(n) = \Theta(n)$
- [ ] $f(n) = \Theta(n^2)$

Question 8:

Second Person:

Using the formal definition of Big-Theta, find and prove the runtime of the expression below. Show your work.

$$f(n) = 3n^2 - |nsin(n)| - 1$$

$\Theta(n^2)$

$3n^2 - |nsin(n)| - 1 \leq 3n^2 - |nsin(n)| - 1 \leq 3n^2 - |nsin(n)| - 1$

$\rightarrow 3n^2 - |n(1)| - 1 \leq 3n^2 - |nsin(n)| - 1 \leq 3n^2 - |n(0)| - 1$

$\rightarrow 3n^2 - n^2 - n^2 \leq 3n^2 - |nsin(n)| - 1 \leq 3n^2$

$\rightarrow n^2 \leq 3n^2 - |nsin(n)| - 1 \leq 3n^2$

$\rightarrow f(n)\ has\ Big - Theta\ runtime\ of\ \Theta(n^2)$

First Person:

Using the formal definition of Big-Theta, find and prove the runtime of the expression below. Show your work.

$$f(n) = 7n^2 - n^2 sin^2(n) - 9$$

$\Theta(n^2)$

$7n^2 - n^2 sin^2(n) - 9 \le 7n^2 - n^2 sin^2(n) - 9 \le 7n^2 - n^2 sin^2(n) - 9$

$\rightarrow 7n^2 - n^2(1) - 9 \le 7n^2 - n^2 sin^2(n) - 9 \le 7n^2 - n^2(0) - 9$

$\rightarrow 6n^2 - 9 \le 7n^2 - n^2 sin^2(n) - 9 \le 7n^2$

$\rightarrow 5n^2 + (n^2 - 9) \le 7n^2 - n^2 sin^2(n) - 9 \le 7n^2$

$\rightarrow 5n^2 \le 7n^2 - n^2 sin^2(n) - 9 \le 7n^2$, for $n \ge 3$

$\rightarrow f(n)\ has\ Big - Theta\ runtime\ of\ \Theta(n^2)$

Question 9:

Both:

Consider the situation where you flip a coin until you get heads. Using the formula for average case runtime in lecture, what is the expected runtime?

constant